

Phase-1: Brainstorming & Problem Definition

Project: Auto Ticket Classification using Flow Designer | Target: u_incident_workflow | Scope: Global

1.1 Operational Context & Baseline Environment

In school and educational IT support environments, students, faculty, and administrative staff submit high volumes of daily service inquiries. Common issues center around classroom projector failures, campus Wi-Fi drops, student/faculty portal credential lockouts, and slow workstation performance.

In the legacy operational flow, every inbound ticket arrives in an unclassified central queue. Helpdesk technicians are forced to manually inspect each record, read the caller's text description, determine the technical category/subcategory, and send an acknowledgment email. This manual triage workflow causes significant operational delays and inconsistent categorization.

Identified Problem Statement: Manual ticket triage creates initial response bottlenecks ranging from 30 minutes to over 4 hours during peak classroom hours, leaving critical learning equipment unattended.

1.2 Brainstormed Solution Concepts & Comparative Evaluation

During initial brainstorming, three architectural approaches were evaluated to address triage inefficiency:

Architectural Approach	Key Advantages	Critical Disadvantages	Decision
Client Scripts & Script Includes	Immediate client-side execution before database commit.	High maintenance overhead; brittle on mobile interfaces; scripts require code releases.	Rejected
Predictive Intelligence / AI	Advanced machine-learning pattern recognition over time.	Requires 30,000+ historical records; high licensing cost; unnecessary for standard taxonomy.	Deferred to Roadmap
Flow Designer Keyword Engine	100% no-code; asynchronous background execution; instant maintainability by administrators.	Requires initial keyword pattern definition.	Accepted (Selected)

1.3 Target Outcomes

- **Zero-Touch Triage:** Automate Category and Subcategory assignment at record insertion.
- **Taxonomic Standardization:** Enforce strict Category-to-Subcategory dependency rules.
- **Immediate Caller Feedback:** Dispatch automated SMTP notifications upon creation.